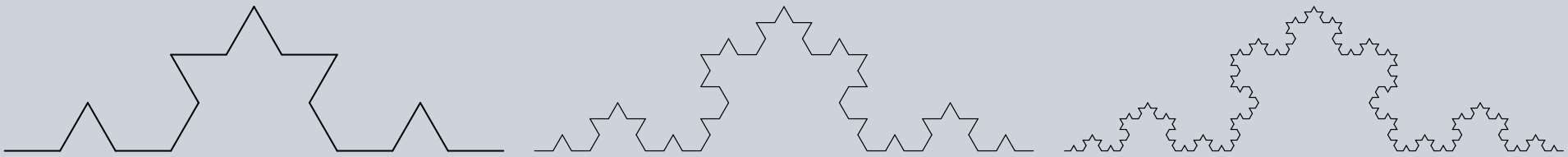
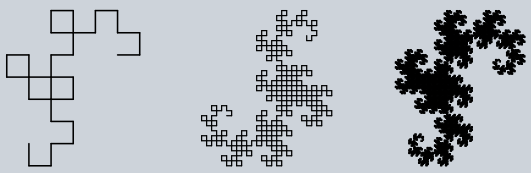
Simple repeated rules create intricate shapes. Within each row the iteration increases from left to right; each panel is resized to make its structure visible.

1 Koch Curve



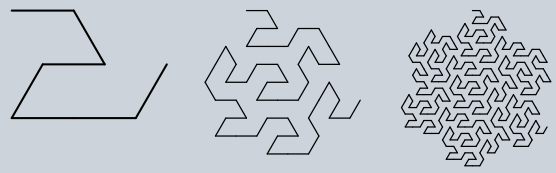
Replace each segment with four segments at one-third scale. The added triangular bump repeats at every level; turn angles are 60 degrees.

2 Dragon Curve



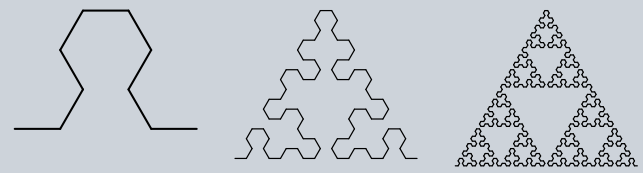
A folding construction becomes repeated right-angle turns. The numbered panels show increasingly detailed iterations, each fitted to its own frame.

3 Gosper Curve



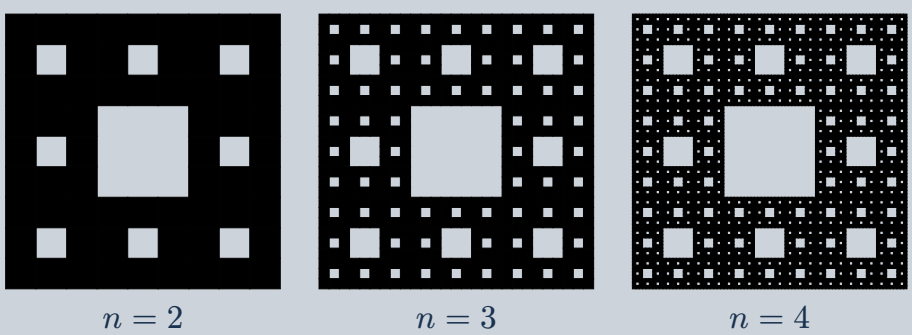
A seven-part replacement on a hexagonal grid builds a denser curve. The two drawing symbols have different replacement rules.

4 Sierpiński Arrowhead



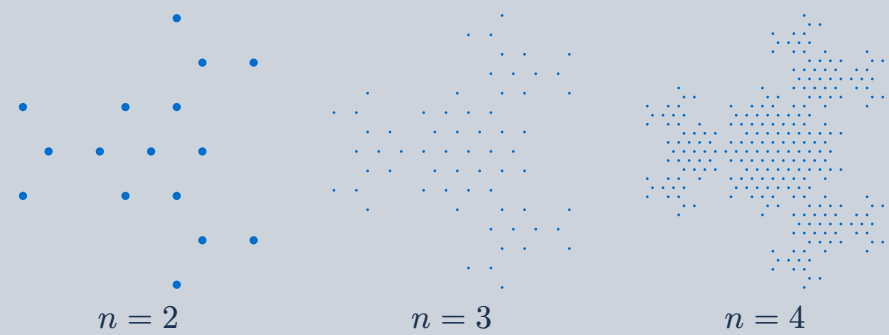
A recursive path of 60-degree turns on a hexagonal grid. It never crosses itself, yet in the limit traces out the Sierpiński triangle.

5 Sierpiński Carpet



Split a square into nine equal cells; remove the middle cell and repeat on the eight survivors. Black marks what remains.

6 Eisenstein



Copy the seed on a triangular lattice, rotate, and expand the arrangement. Here the marks are points rather than a continuous turtle path.

Same rule, finer detail. n is the iteration index, with the seed convention shown by each construction. Panel resizing hides changes in absolute size; compare structure rather than printed length.